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Digital forensics in law enforcement: A case study of LLM-driven evidence analysis

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ABSTRACT

The advent of digital technology and the ubiquity of mobile devices in today's society has led to a significant increase in the importance of mobile forensics in criminal investigations. Responding to the escalating volume and complexity of data due to enhanced smartphone capabilities and pervasive messaging apps, law enforcement agencies face challenges in data analysis. This study explores improving investigative efficiency through LLM-driven analysis of text from mobile messenger communications. We have conducted experiments on anonymized data collected from real crime scenes by employing three state-of-the-art LLM models, namely GPT-4o, Gemini 1.5 and Claude 3.5. The study focuses on optimizing model performance by employing prompt engineering, interpreting expressions embedded with hidden meanings such as slang, and contextually inferring ambiguous word usage. Finally, model performance is quantitatively evaluated using metrics such as precision, recall, F1 score, and hallucination rate.

1. Introduction

With the advancement of digital technology and mobile devices that permeate all aspects of daily life, the importance of mobile forensics in criminal investigations has increased significantly (Alshameri et al., 2024). As smartphone storage capacity increases and various messenger apps become widespread, law enforcement agencies are facing an era in which the volume and complexity of data to be analyzed is increasing exponentially (Sjöstrand, 2020). In this context, this study proposes a method to improve investigative efficiency by applying LLM technology to the analysis of text from mobile messenger messages.

The current investigative practice of identifying crime related keywords (such as money transfers, transaction objects, locations) through keyword searches is a prevalent method. This study specifically examines through experiments, the extent to which LLMs can contribute to the extraction of key messages related to criminal contexts through message context analysis. The experiments were conducted using anonymized data collected from actual crime scenes, and three state-of-the-art LLM models – GPT-4o, Gemini 1.5, and Claude 3.5 – were used to identify the most effective model in the field of mobile forensics.

In this process, efforts were made to maximize the models' identification performance through prompt engineering, interpreting expressions with hidden intentions such as slang, and inferring the usage of words through context when ambiguous expressions appear. Finally, the models' performance is quantitatively evaluated through metrics such as Precision, Recall, F1 score, and Hallucination Rate.

1.1. Contribution

Our research makes several significant contributions to the field of digital forensics, particularly in the area of mobile device investigations using real-world police data. Through comprehensive analysis of actual criminal investigations, we propose a novel mobile forensics framework that enhances the efficiency and accuracy of digital evidence analysis through AI technologies.

- We used over 140,000 real-world forensic messages from mobile phones seized in drug-related investigations by the Korean National

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Police Agency, with Ground Truth established by experienced investigators. This practical approach validates the real-world applicability of our proposed methodology in actual investigation environments.
- We developed a novel framework that integrates AI and data visualization technologies to significantly enhance both the efficiency and accuracy of digital crime investigations, providing investigators with more powerful and intuitive tools for evidence analysis.
- Our research presents detailed experimental designs and results, along with a critical discussion of limitations and future research directions, contributing to the methodological advancement of mobile forensics investigation techniques.

2. Related works

The field of digital forensics has continuously evolved with the artificial intelligence(AI) technologies, leading to numerous research initiatives focused on developing sophisticated tools and methods to enhance forensic analysis and criminal investigations.

Mukkamala and Sung (2003) pioneered AI integration in network forensics through offline intrusion analysis, while Case et al. (2008) established automated correlation analysis between digital evidence. Hoelz et al. (2008) advanced this field by developing multi-agent systems and case based reasoning approaches.

Scanlon et al. (2023) evaluated ChatGPT's potential in digital forensics across six domains, highlighting its utility as a supplementary tool while noting limitations requiring expert oversight. Oh et al. (2024) introduced volGPT, successfully implementing prompt-based LLMs for explainable memory forensics triage with high accuracy in ransomware detection.

Fayyaz et al. (2024) developed a hybrid AI framework for analyzing vehicle infotainment system data, while Yang et al. (2020) demonstrated CNN's effectiveness in image source forensics. Vasilaras et al. (2024) evaluated AI-based mobile forensics tools, emphasizing transparency and ethical considerations. Afchar et al. (2018) and Orozco et al. (2020) advanced video forensics through deepfake detection and post-processing analysis respectively.

In recent developments, Qamhan et al. (2021) applied CRNN models to audio forensics, while Rughani (2017) proposed a comprehensive AI-driven forensics framework integrating collection, analysis, and reporting stages.

These studies collectively demonstrate AI's crucial role across digital forensics domains, emphasizing both its potential and the importance of maintaining legal validity and reliability.

3. Research problem and proposals

Law enforcement agencies always operate under time constraints. For example, if a criminal is tracked down, apprehended, and arrested in South Korea, and you want to keep him or her in custody, you have 36 hours to apply for an arrest warrant that more specifically defines the charges against the suspect, and includes evidence that demonstrates the suspect's risk of fleeing the country, such as fear of destruction of evidence or fear that the suspect will flee again. Furthermore, even after this process, there is a time limitation as investigators must gather evidence and obtain suspect statements within 10 days before transferring the case to the prosecution.

Meanwhile, the volume of data that law enforcement agencies must analyze to prove criminal charges is continuously expanding. Analyzing electronic information stored on suspects' mobile phones has become mandatory rather than optional. As phone storage capacities increase, the total volume of data that investigators must examine continues to grow (Fayyaz et al., 2024).

Particularly, with the development of various mobile messengers (Telegram, WeChat, Line, etc.), most people now communicate via mobile messengers rather than phone calls. Mobile messengers have become the most crucial communication channel in modern digital society, offering not only the ability to communicate through photos and videos but also various convenience features (money transfers, purchases, voting, etc.). This study aims to discuss methods for efficiently exploring such mobile messenger data within the aforementioned time constraints and extracting and analyzing information relevant to criminal charges.

One of the most widely used methods by law enforcement agencies to search for crime-related electronic information within mobile messengers is keyword searching. For example, when conducting forensics on a drug suspect's phone, investigators extract message data through various mobile forensic tools and filter chats containing specific keywords like "drugs" for investigation.

Law enforcement agencies face major challenges when dealing with such extracted data. It requires us to put an enormous effort and substantial time to determine whether the extracted messages are genuinely related to certain crimes or a message refers to merely metaphorical use of "drug" word (e.g., describing a song as "addictive as drugs"). This requires investigators to analyze extensive amounts of adjacent messages to understand the context.

The key issues in mobile forensics, particularly in analyzing messenger data, can be summarized as follows:

- Digital forensics procedures must be handled in a swift and accurate manner given such strict time limits.
- Solutions are needed to address the problem of investigators having to manually verify the context in which the message related to crime is located.

To address these key issues, this study proposes the framework below: Utilizing SoTA(State of The Art) LLM models to verify message context - examining the possibility of applying them to digital forensics through various experiments with the actual data provided by law enforcement agencies.

4. Framework

4.1. Framework overview

We propose a framework that can help investigators to efficiently detect criminal clues to prove criminal offenses during digital forensics, thereby reducing time needed. The overall architecture is shown in Fig. 1.

This starts by acquiring mobile device data using some forensic softwares from mobile devices such as smartphones, SD memory cards, and IoT devices. This kind of software enables us to download the result file as preferred formats such as excel worksheet, csv, text, etc. Those output files from the software include evidence information, message history, and some metadata which is useful in the investigation process.

In the next step, we should make those output files applicable for feeding LLMs by pre-processing them.

And then we utilize LLMs, such as ChatGPT, Claude, and Gemini, reading all the context messages fed and determining its relevance to the case. At this point, we apply prompt engineering techniques to finely tune the prompts, helping LLM to extract important and accurate information to deliver optimal results.

Prompt Engineering is used here to craft or refine the input prompts submitted to the LLMs, based on the structured data and specific investigative needs. This includes rephrasing, contextual restructuring, or adjusting instruction strategies to optimize the LLM's output.

The output from the LLM is subsequently reviewed through the investigator's judgment process, a critical decision point in the framework. Here, the investigator evaluates whether the extracted information is appropriate, relevant, and sufficiently complete for the case at hand. If the output meets the investigative requirements, the process ends. However, if the results are unsatisfactory—e.g., incomplete, ambiguous, or misleading—the investigator initiates a re-extraction process.

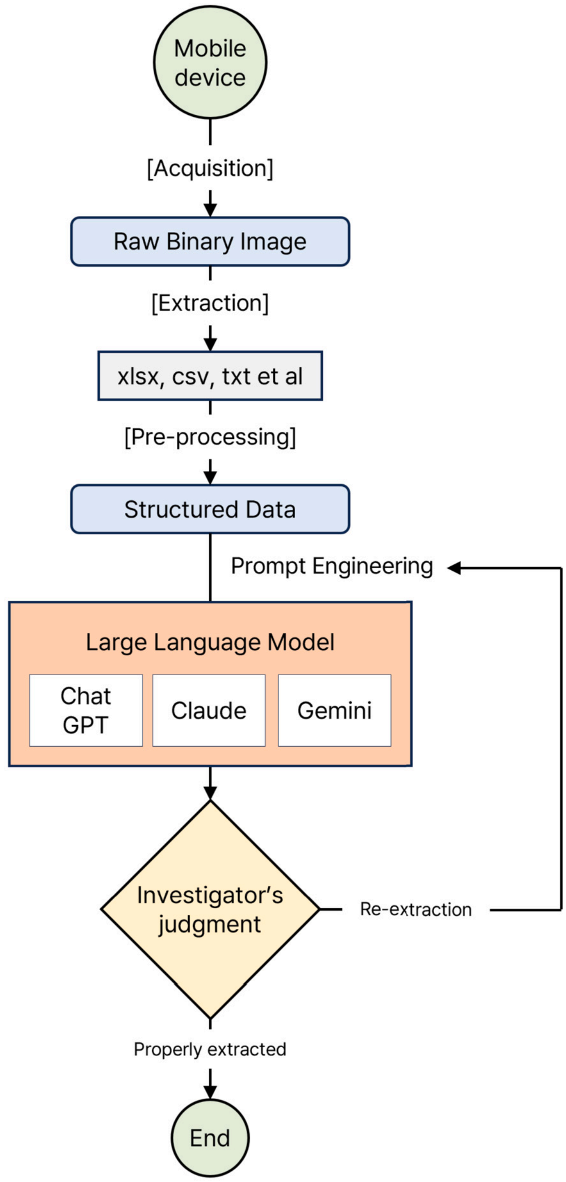

Fig. 1. Overview of LLM-driven Evidence Analysis Framework.

In the re-extraction branch, Prompt Engineering is revisited not as part of the Structured Data step, but as an independent optimization step applied after the structured data has already been prepared. This is because Prompt Engineering directly influences how LLMs interpret the input and extract useful content, rather than altering the underlying data structure. The process thus loops back from the investigator's judgment to Prompt Engineering, which in turn sends the refined prompts back to the LLM.

This explains why Prompt Engineering is positioned alongside the LLM module rather than under Structured Data—it is more logically associated with optimizing LLM interaction than modifying the structured input itself.

This loop continues iteratively until the investigator determines that the extraction is satisfactory and the information can be reliably used for evidentiary purposes.

4.2. Acquisition & extraction software

The analysis of smartphone communication records is conducted using a two-step procedure. The first step is the physical extraction of data from the device, followed by the carving of files into excel worksheet format from the extracted binary data. While this process can be accomplished using open-source software, it typically requires more effort than using commercial tools. In this study, we employ MD-NEXT and MD-RED, developed by GMDSOFT, a Korean company specializing in mobile forensic solutions. MD-NEXT is a professional mobile data acquisition tool that supports logical, file system, and physical extraction from a wide range of smartphones, including Android and iOS devices. MD-RED is its companion tool for comprehensive data analysis, enabling investigators to review messages, call logs, app data, and metadata in a structured, user-friendly interface. These tools together provide an end-to-end solution for mobile forensic investigations from data extraction to in-depth analysis and reporting.

A summary of the specific details regarding the software and smartphone models used in this study can be found in Table 1.

Table 1
Software and device specifications used in the study.

Index	Device	Version	Tool
1	iPhone 11 XS	iOS 15.6.1	MD-RED 3.10.16.2360
2	Samsung Galaxy Note 10+	Android 10.2.2	MD-NEXT 2.0.14.2259
			MD-RED 3.11.10.2964

4.3. Pre-processing

After the anonymization, the whole dataset is too large to build ground truth or to directly feed it to LLMs. In addition, most messages are sent in pieces. So we performed pre-processing to tackle these issues and to make the dataset proper to be sent to LLMs.

Firstly, we make keyword searches with certain terms, in this study we used 'drug', and we have to get some additional messages from the chat rooms or chat opponents to ensure the message context. Thereby we get sufficient message sets which are appropriate to present to professionals or LLMs.

4.4. Prompt engineering

Prompts provide essential information for a model to accurately understand a given problem and perform appropriate analysis, and their structure and specificity are key determinants of the quality of a model's response (Brown, 2020).

The prompts in our study are organized as follows: We specify the role of the model as a "mobile forensic expert," provide background information about the situation under investigation, and instruct the model to identify and extract text relevant to the alleged crime. For example, the prompt might be written in the following form:

f"{target_text} \n\n

Determine whether the above text is related to drug crimes. For your reference, I will provide you the target message and 40 additional messages from before and after the target message in a key-value format. Key is a nickname and value is the content of the conversation.

Imagine that you're an investigator looking into messages from the seized phones, and you have to identify the target message's relation to the drug crime with clear and reasonable criteria. Please give me an output of 1 or 0, 1 refers to Positive and 0 refers to negative. You should mainly focus on whether the people in the conversation are directly involved in drug crimes (trafficking, drug use, or drug distribution, etc.). So just mentioning the offense of celebrities, politicians is negligible.

On the basis of criteria above, make an overall judgment on whether or not the message and sender are involved in specific drug crimes and return the result. Below is the chat history for your reference. \n\n{chat history}"

Table 2
Model specifications comparison.

Model	Max. Input Tokens	Max. Output Tokens
GPT-4o	128,000	16,384
Gemini 1.5 Pro	2,088,959	8,192
Claude 3.5 Sonnet	200,000	8,192

Table 3
Data policies of LLM providers.

Model Name	Policy
GPT-4o (OpenAI)	(Enterprise) Input data is not used for training purposes. (Free, Plus) Can opt out not to use input data as a training purpose through data controls.
Gemini 1.5 (Google)	Does not use data (prompts or responses) to train its models without your permission.
Claude 3.5 (Anthropic)	(Anthropic API) Input data is not used for training purposes.

This prompt leads the model to act as an expert in investigation and, given the background information, to analyze the context in the message data and accurately extract important information related to the alleged crime.

However, in the case of the GEMINI API, when the 'instructions' and 'input text' parts are directly connected, there exists a problem that the model forgot the instructions and focused on the 'input text', which prevents the model from fulfilling its intended role and leads to the model focusing on the details of the input text and failing to identify important information related to the alleged crime.

To address this issue, we implement the Sandwich Defense approach (Schulhoff) to ensure that the model does not forget instructions. This approach changes the structure of the prompt into "instruction," "input text," and "reaffirmation of instruction" to help the model repeatedly recognize and retain the instruction. Here, reaffirmation of instruction means telling the model one more time, "I'll say it again, you should output a 1 or 0 if this is related to the crime."

4.5. Contextual grasp with LLM

Understanding the context of messages from seized mobile devices is essential for investigation procedures. Since existing methods have limitations in effectively interpreting tabular data, we propose using SoTA LLMs, GPT-4o, Gemini 1.5, and Claude 3.5, to sift through pre-processed data.

Comparing the specifications of the number of input and output tokens of the models used in this study, each model is characterized as shown in Table 2.

Given that these models generate answers based on our input data, issues related to privacy and copyright infringement can be raised (Yang, 2023). In this perspective, we cannot exclude the possibility that the sensitive data may be leaked to the outside.

The three LLMs mentioned above have their own data policies about data leakage, reuse of input data as training purpose, data ownership, processing methods, and privacy measures. Further details can be found in Table 3.

Although we anonymized the original data in a way shown in Section 5.2.2, we also found the privacy policy of each of the LLMs which stated that they do not use input data for training models when using commercial plans, especially via APIs.

5. Experiments

5.1. Environment

The computing environment used in this study is a high performance workstation specifically configured to support intensive forensic data analysis tasks. The workstation is equipped with a 24-core Intel(R) Xeon(R) Gold 5220R CPU running at 2.20 GHz, providing powerful computational power for parallel processing. This configuration enables efficient processing of multi-threaded applications and large datasets. The system is equipped with 64 GiB of RAM, which ensures smooth performance without memory-related bottlenecks during complex calculations and data processing tasks. The workstation also features an NVIDIA GeForce RTX 2060 GPU, which has the ability to accelerate tasks such as training machine learning models and analyzing forensic data.

5.2. Dataset

5.2.1. Preview of the dataset

We acquired smartphones seized by law enforcement agencies from real world drug cases. Then we extract message data and export it into excel worksheet format as mentioned in Section 4.2. The whole message dataset used in this study consists of 31 columns and 142,214 rows. The main columns are shown in Table 4. We translated all our data for the experiment into English.

5.2.2. Anonymization

We anonymized personal information such as names, addresses and phone numbers to ensure privacy and to proceed for further research.

We use NER (Named Entity Recognition) to anonymize names and generate random numbers to replace contacts. In the case of emails, we replace all the usernames with the same amount of '#'. It is also important to consider the possibility of generating the same number as existing contact when anonymizing contacts. Therefore, we generated random numbers within the range of middle digit numbers which are not being used in Korea to minimize the possibility of duplication with existing contacts.

5.2.3. Pre-processing

We applied a series of data processing methods mentioned in our framework to real world drug cases. We extract the first 20 rows and the last 20 rows of the corresponding chat, and combine them into a single data set. In this way, we extract 200 data points from the first drug case and 200 data from the other drug case (we obtained data from two real-world drug cases). As a result, we get 400 excel workbook files where the keyword matching messages are presented with the contexts. However, out of a total of 400 datasets, one dataset was not successfully classified by the LLM model, so the final 399 datasets were used in the experiment in Section 5.4.

5.2.4. Ground truth & cross validation

To evaluate the performance of our framework, we need to label the dataset mentioned in Section 5.2.3 whether they are truly related to drug crimes or they are just including slang or some metaphorical use of certain words. We asked two experienced investigators for labeling each single dataset whether it is actually related to drug crime or simply referring to figure of speech. Specifically, they were given target data with 20 preceding and 20 following sentences and labeled it 1 if the data was related to drug crimes and 0 otherwise.

We then measured Cohen's Kappa (Cohen, 1960) to determine the quality of this annotation. Cohen's Kappa is a statistical measure used to assess the degree of agreement between two raters (or annotators) in classifying items into mutually exclusive categories.

The formula for Cohen's Kappa (κ) is:

$$\kappa = \frac{P_o - P_e}{1 - P_e} \tag{1}$$

Where:

- Observed Agreement (P_o) = Total number of agreements / Number of observations

Table 4
Main columns of the dataset.

Column Name	Description	Example
App	Message application name	'Message', 'Telegram', 'Kakao Talk', 'Instagram'
Status	Message status	'Active', 'Deleted', 'Status'
Type	Message type	'Received', 'Sent', 'System Message', 'Chat Room Setting'
Content	Actual text content	'Hello, sales will be made in order of arrival'
Chat Room	Unique chat room number	242, 10092911026
Group	Whether group chat	'True'
Sender Number	Sender's phone number	'01012345678'
Sender	Sender's name	'Ming Ming', 'Comeboy'
Receiver	Receiver's name	'Uni', 'Leica'
Receiver Number	Receiver's phone number	'114', '01011111111'
Creation Time	Message creation time	2023-05-08 10:11:16
Message ID	Unique message identifier	'9D36093A-C864-E902-62E2-D695E5746403'
Account	Account information	'+821004704241'

- Expected Agreement (P_e) $= \sum_{i=1}^{k} P_{i+}P_{+i}$
- P_{i+} is the percentage of items assigned to category i by annotator 1
- P_{+i} is the percentage of items assigned to category i by annotator 2
- k is the number of categories

In our experiment, the observed agreement (P_o) was 0.87, and the expected agreement by chance (P_e) was 0.50, resulting in a Cohen's Kappa value of 0.74. According to Landis and Koch (1977), a value between 0.61 and 0.80 indicates substantial agreement, demonstrating the high quality of our annotation.

5.3. Comparison metric

In our study, we use Precision, Recall, F1-score, and Hallucination Rate to evaluate how well large language models can discriminate between actual drug crimes and figurative expressions in message data. For example, a sentence such as "This music is as addictive as drugs" is not relevant to the crime, but a sentence such as "Let's trade drugs for this price" is directly relevant to the crime. According to Powers (2020), Precision refers to the percentage of items positively predicted by the model that are actually correct, and is used to evaluate the ability of the model to accurately identify whether a particular sentence is text actually used in a drug crime or not.

The formula for Precision is:

$$Precision = \frac{TP}{TP + FP} \tag{2}$$

TP(True Positive) is a correctly identified positive case. FP (False Positive) is an incorrectly identified negative case as positive. TN(True Negative) is a correctly identified negative case. FN(False Negative) is an incorrectly identified positive case as negative.

Recall is the percentage of sentences that were actually used in a drug crime that the model correctly identified, and is used to evaluate the ability to not miss important crime-related information.

Recall is calculated as:

$$Recall = \frac{TP}{TP + FN} \tag{3}$$

F1-score is a metric that evaluates the balance between Precision and Recall, and measures the overall performance of the model through a harmonized average of the two metrics. In particular, it is used to comprehensively evaluate how well the model understands the drug-related context and effectively categorizes the relevant sentences.

The F1-score is calculated as:

$$F1\text{-}score = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{4}$$

Hallucination refers to the phenomenon where a large language model (LLM) produces information that does not actually exist or incorrect information (Maynez et al., 2020). These errors can occur during the process of inference or deduction from the data on which the model was trained, and can be particularly problematic in fact-based tasks. In general, hallucinations can degrade the reliability and accuracy of a model, making them an important metric when evaluating model performance. We define a hallucination as a case where the LLM incorrectly judges the ground truth to be 1 when it is 0. Based on this definition, we calculate the Hallucination Rate as the rate at which the model produces incorrect outputs that do not match the actual data, and use this metric to assess whether the model's output is reliable in an investigation.

The Hallucination Rate is calculated as:

$$Hallucination\ Rate = \frac{FP}{TP + FP} \tag{5}$$

5.4. Experiment results

We compared the results from the LLMs to the ground truth annotated by the investigators as Table 5.

- GPT-4o performs well on all metrics, especially recall (0.914) and F1 score (0.899). However, it does have a moderate hallucination rate (0.116).
- Gemini 1.5 is excellent when it comes to accuracy and hallucination-free, but has a low recall (0.782).
- Claude-3.5 has the lowest accuracy (0.794) and the highest hallucination rate (0.206).

Given that no model is superior in every metric, we introduce the majority vote system. In this system, the final result is a value of 1 (crime) if at least two of the above three models judge it to be 1, and 0 otherwise. This counterbalances the tendencies of each model to produce a more stable decision, resulting in high precision (0.944), moderate recall (0.835), and very low hallucination rate (0.056).

In our experiments with the APIs of GPT-4o, Gemini 1.5, and Claude-3.5, we set the temperature = 0 parameter in all models to get as consistent results as possible. Temperature = 0 parameter makes the model more deterministic by reducing randomness in responses (Peeperkorn et al., 2024).

We present graphic performance comparison in Fig. 2.

6. Conclusion & future work

6.1. Summary

This research presents a comprehensive framework for digital forensics, aiming to enhance the efficiency of investigators in sifting through crucial evidence for proving alleged crimes, thereby reducing investigation timeframes. The framework encompasses data extraction from diverse mobile devices using forensic software, preprocessing the obtained data, and applying prompt engineering to comprehend the context. This approach proves invaluable when rapid data analysis is required, facili-

Table 5
Overall Performances of each model.

Model	TP	FN	FP	TN	Precision	Recall	F1 score	Hallucination rate	Processing time
GPT-4o	222	21	29	127	0.884	0.914	0.899	0.116	25 m 33 s
Gemini 1.5	190	53	0	156	1	0.782	0.878	0	8 m 31 s
Claude-3.5	208	35	54	102	0.794	0.856	0.824	0.206	65 m 59 s
Majority Vote	203	40	12	144	0.944	0.835	0.886	0.056	-

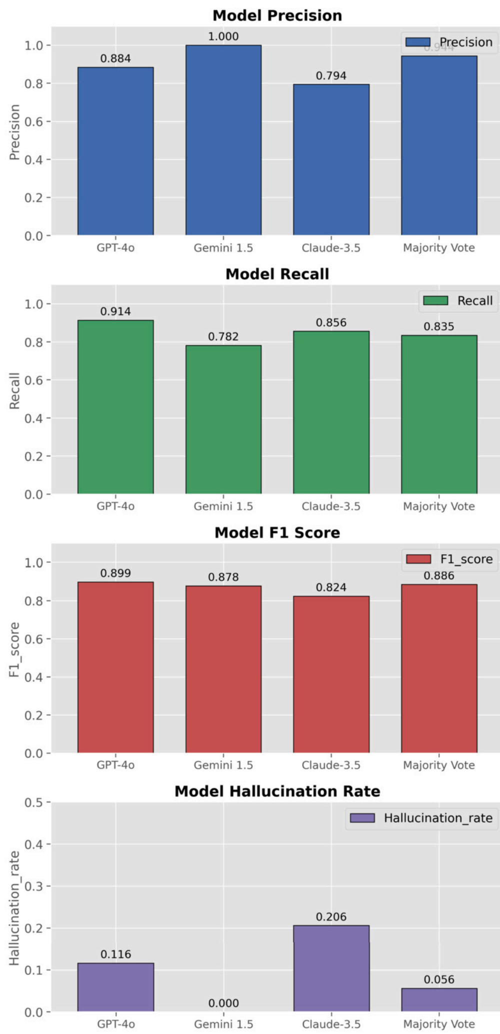

Fig. 2. Model Performance Comparison.

tating the intuitive identification of significant clues within constrained time limits.

When analyzing how effectively each model extracts information relevant to the alleged crime in real-world investigative situations, GPT-4o performed well, with the highest overall recall and F1 score. In contrast, Gemini had a lower recall but fewer false positives, and Claude had relatively low precision. The results of this performance analysis of each model are important to understand the strengths and limitations of each model, and we figure out that the majority voting method compensates for the conflicts between the models to produce stable and reliable results.

6.2. Limitations and future works

Although we discovered the usefulness of employing several SoTA LLMs in the crime investigation process, we still have some challenges in extracting information related to the case from mobile devices.

Firstly, we keyword-searched for sentences containing the certain word “drugs”, and created a set of messages by grouping preceding and following text messages. We then asked LLMs to judge their relevance to drug crimes based on the context of the set of messages. However, this method excludes the possibilities of crime-related messages written in metaphorical terms or indirect terms. Therefore, future research should move beyond simply searching for certain keywords. Instead, it should focus on embedding the entire dataset. This approach will enable a more sophisticated analysis of semantic similarity. By doing so, we can better compare messages with target words or sentences. We expect this method to overcome the limitations and extract more accurate information that takes into account more complex contexts.

Secondly, we experienced that our suggested framework is subject to each of the policies of LLMs. Sometimes they offer only limited responses to topics such as crime, violence, and drug dealing for user protection and safety. In addition, these LLMs prevent themselves from learning continuously updated laws or confidential investigative data (Kim et al., 2024), and they pose security concerns due to the risk of exposing sensitive information from criminal cases while using the API (Liu et al., 2023). To overcome these constraints and provide more accurate and secure customized responses for Law Enforcement Agencies, we should consider implementing a local model which is relatively small sized or customized for special fields of knowledge.

We can utilize a Retrieval-Augmented Generation (RAG) model as an alternative to purely generative models, which rely solely on pre-trained knowledge. By combining RAG with a Small Language Model (SLM) like a smaller-parameter version of LLaMA, we can achieve better performance by enhancing factual accuracy, reducing hallucinations, and improving domain-specific responses through real-time retrieval (Gao et al., 2023). RAG models work by searching for relevant documents in real-time to add information that can be referenced when generating answers, rather than relying solely on pre-trained knowledge.

Thirdly, with respect to the majority voting method used in this experiment, the majority voting method provides high precision and moderate recall, as mentioned in Session 5.4, but has the disadvantage of requiring more time and money to provide data and make judgments for both GPT-4o, Gemini 1.5, and Claude-3.5 models.

Last but not least, prompt engineering has several limitations when applied to sensitive and context-dependent domains such as criminal investigation. In particular, the structure and wording of the prompts directly affect the performance of the model, and even simple structural changes can lead to significant changes in the generalization performance of the model (Schick and Schütze, 2020; Zhao et al., 2021).

When using the Gemini api, we found that the ‘input text’ directly followed by the ‘instructions’ caused the model to forget about the instructions and focus on the input text, which resulted in the model failing to identify important crime-related information. We adopted the Sandwich Defense approach to settle this down as we mentioned in Section 4.4. However, the problem remains.

For future work, we suggest an adaptive prompting framework to compensate for these issues, in which the prompts can be dynamically adjusted according to the complexity and context of the input data. For example, there is a need to design hierarchical prompts that help the model understand the context step by step instead of processing all the details at once (Schick and Schütze, 2020). In addition, we hope to combine prompt engineering with few shot learning to design prompts with representative examples, which will allow the model to generalize in complex and diverse situations (Gao et al., 2020).

CRediT authorship contribution statement

Kyung-Jong Kim: Writing – original draft, Validation, Project administration, Investigation, Data curation, Conceptualization. **Chan-Hwi Lee:** Writing – original draft, Validation, Software, Investigation, Data curation. **So-Eun Bae:** Writing – review & editing, Visualization, Validation, Investigation, Data curation. **Ju-Hyun Choi:** Writing – review & editing, Validation, Investigation, Formal analysis, Data curation. **Wook Kang:** Writing – review & editing, Validation, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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